

● HARD

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

09

Q1

ADVANCED MATH

The function f is defined by $f(x) = a\sqrt{x+b}$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the point $(-24, 0)$, and $f(24) < 0$. Which of the following must be true?

- A. $f(0) = 24$
- B. $f(0) = -24$
- C. $a > b$
- D. $a < b$

Q2

ADVANCED MATH

The area of a rectangular banner is 2,661 square inches. The banner's length x , in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

- A. $0 = x^2 - 24x - 2,661$
- B. $0 = x^2 - 24x + 2,661$
- C. $0 = x^2 + 24x - 2,661$
- D. $0 = x^2 + 24x + 2,661$

Q3

ADVANCED MATH

$$fx = 2722^x$$

The function f is defined by the given equation. If $hx = fx - 4$, which of the following equations defines function h ?

- A. $hx = 172^x$
- B. $hx = 682^x$
- C. $hx = 27216^x$
- D. $hx = 2728^x$

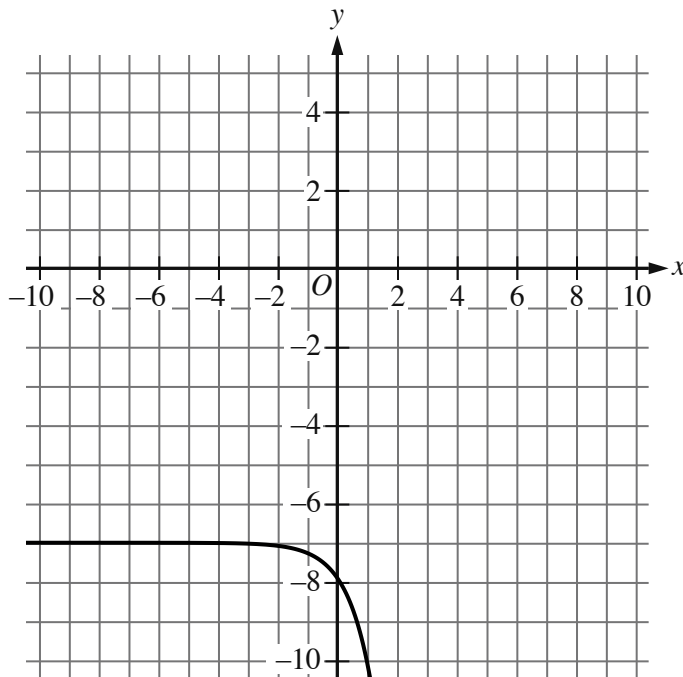
Q4

ADVANCED MATH

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$.

What is the value of ab ?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 54
- D. 60



- Moving from left to right, the curve passes from quadrant 3 to quadrant 4.
- As x decreases, the curve approaches line y equals negative 7.
- The curve passes through the following approximate points:
 - (negative 1 comma negative 7.3)
 - (0 comma negative 7.9)

The graph of $y = fx$ is shown, where $fx = ab^x + c$, and a , b , and c are constants. For how many values of x does $fx = 0$?

- A. Three
- B. Two
- C. One
- D. Zero

An auditorium has seats for 1,800 people. Tickets to attend a show at the auditorium currently cost \$ 4.00. For each \$ 1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- A. 4
- B. 7
- C. 14
- D. 18

Q7

GRID-IN ADVANCED MATH

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y . For what value of x does the value of y reach its minimum?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f ?

A. $f(x) = x + 44^2$

B. $f(x) = x + 176^2$

C. $f(x) = 176x + 44^2$

D. $f(x) = 176x + 176^2$

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function $f(x) = \frac{1}{9}x^2 - 7x + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A. The metal ball's minimum height was 3 inches above the ground.
- B. The metal ball's minimum height was 7 inches above the ground.
- C. The metal ball's height was 3 inches above the ground when it started moving.
- D. The metal ball's height was 7 inches above the ground when it started moving.